



Biodiversity → Information

HADC! (HARPER ADAMS DATA CLUB)



Ed Harris

2025-11-03



What is biodiversity?



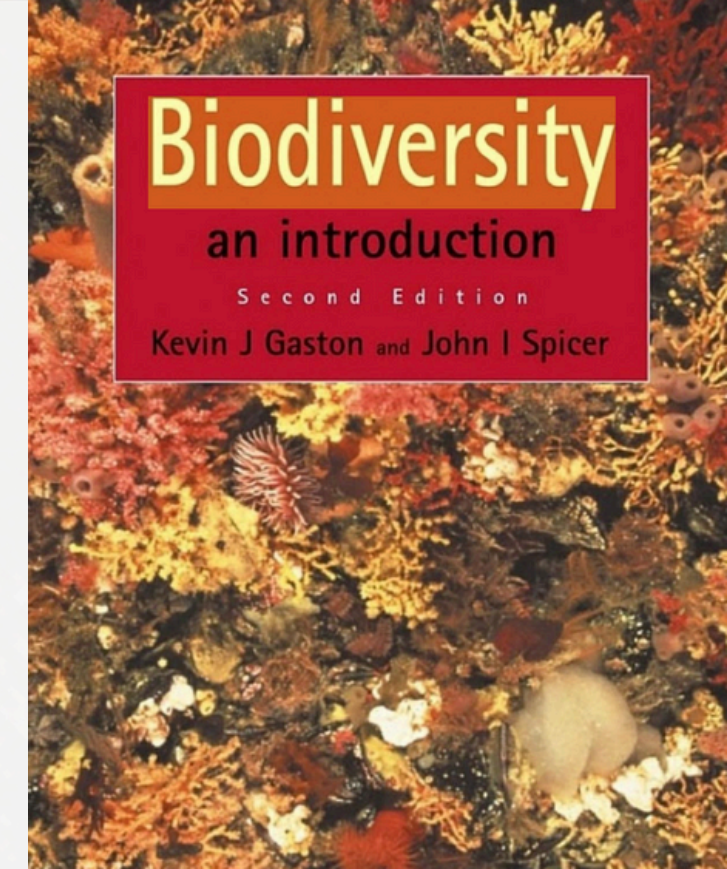
- **SPECIES (RICHNESS)**
COUNTING SPECIES

- **DISTRIBUTION OF SPECIES (EVENESS)**
SPECIES TEND TO BE EVENLY DEISTRIBUTED VERSUS
DOMINATED BY A FEW

- **FUNCTIONAL**
BIODIVERSITY OF HABITAT FUNCTIONS LIKE WETLANDS,
OLD FORESTS, GRASSLAND, ETC.

- **GENETIC**
LITERALLY GENETIC VARIATION

- **PHYLOGENETIC**
SPECIES GROUPS THAT ARE UNIQUE EVOLUTIONARILY



— What is biodiversity?



OFTEN WE ARE TALKING ABOUT

**OCCURENCE DATA FOR INDIVIDUAL
SPECIES OR TAXA**



— Why care about biodiversity?

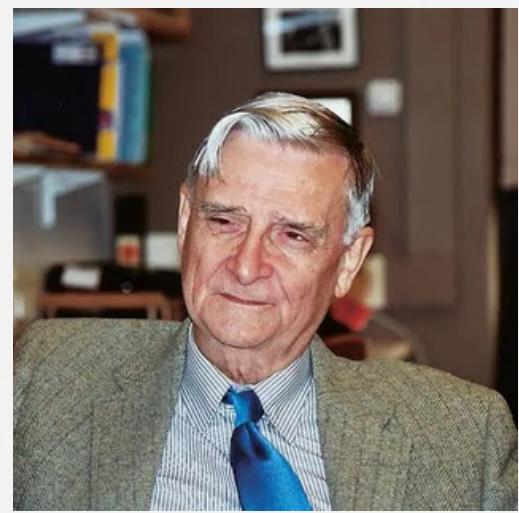
BIOPHILIA, THE HYPOTHESIS THAT HUMANS EVOLVED “IN NATURE” AND THE VAST REDUCTION OF NATURAL ENVIRONMENTS HAS CAUSED MANY PHYSICAL AND PSYCHOLOGICAL HEALTH ILLS

ECOSYSTEM SERVICES LIKE POLLINATION, PEST CONTROL, FLOOD STABILISATION

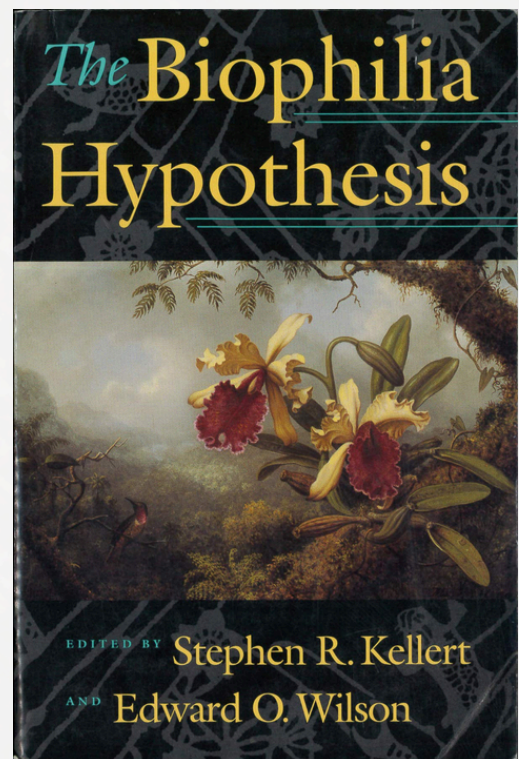
ETHICS SUCH AS THE BELIEF ALL SPECIES DESERVE TO WAX OR WANE

ECONOMIC VALUE FOR TOURISM, FUTURE CROPS OR PHARMACEUTICALS

ENVIRONMENTAL CONCERNS SUCH AS BREAKDOWN OF NATURAL BIOLOGICAL SYSTEMS FOR SOIL, THE OCEANS, ETC.



E. O. Wilson



How to measure biodiversity?

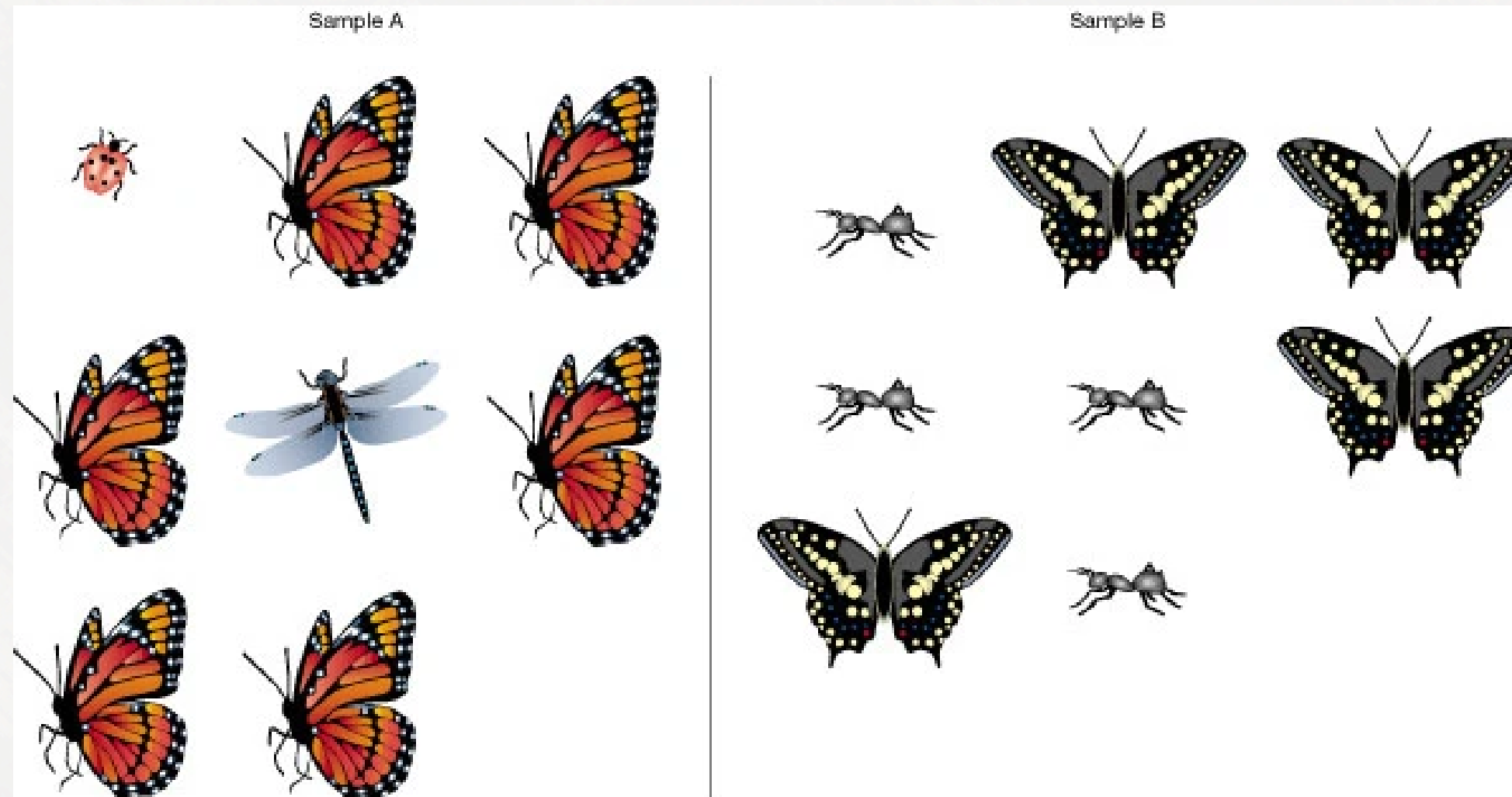
Review Article | Published: 11 May 2000

Getting the measure of biodiversity

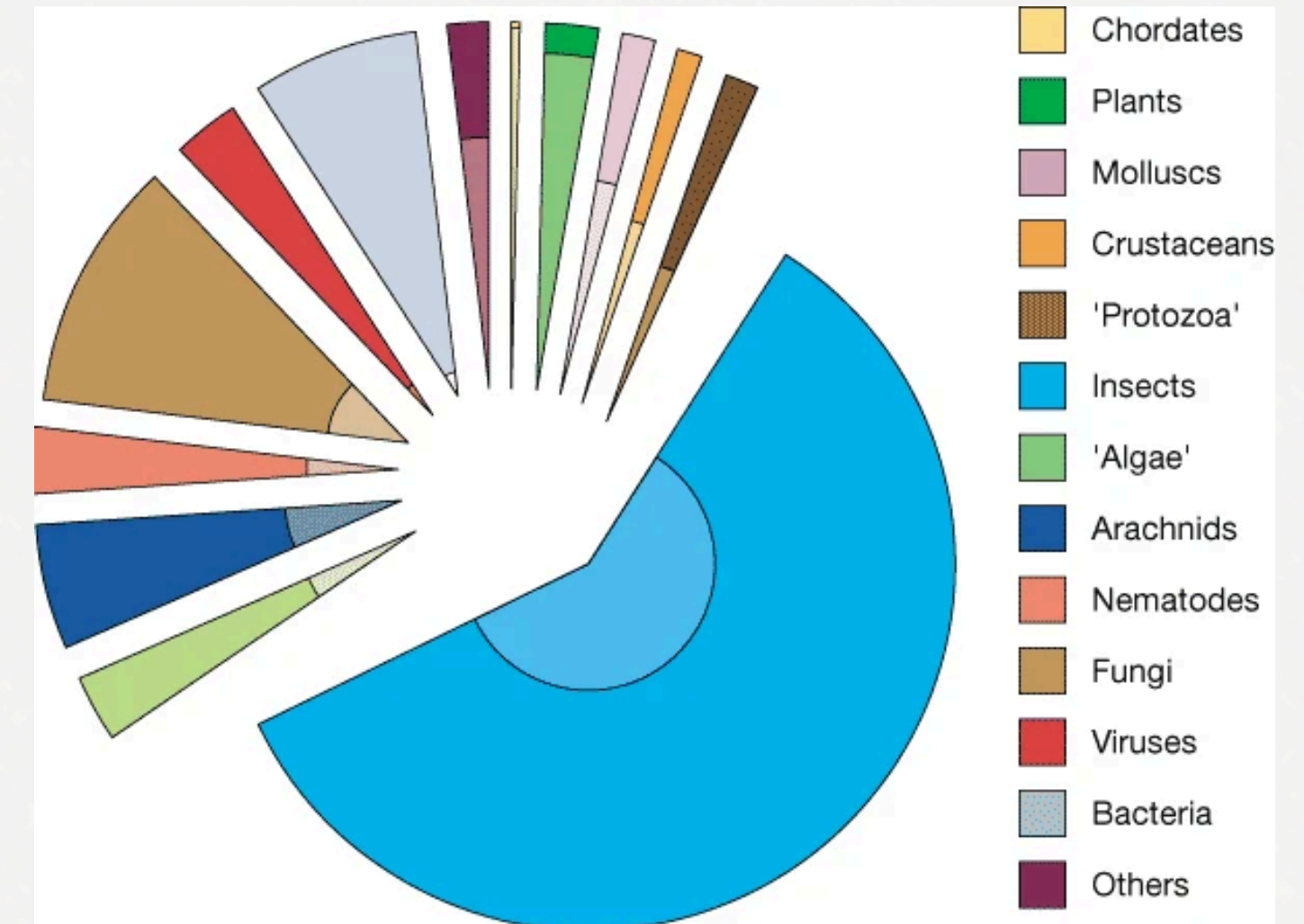
[Andy Purvis](#) & [Andy Hector](#)

[Nature](#) **405**, 212–219 (2000) | [Cite this article](#)

44k Accesses | **1088** Citations | **63** Altmetric | [Metrics](#)



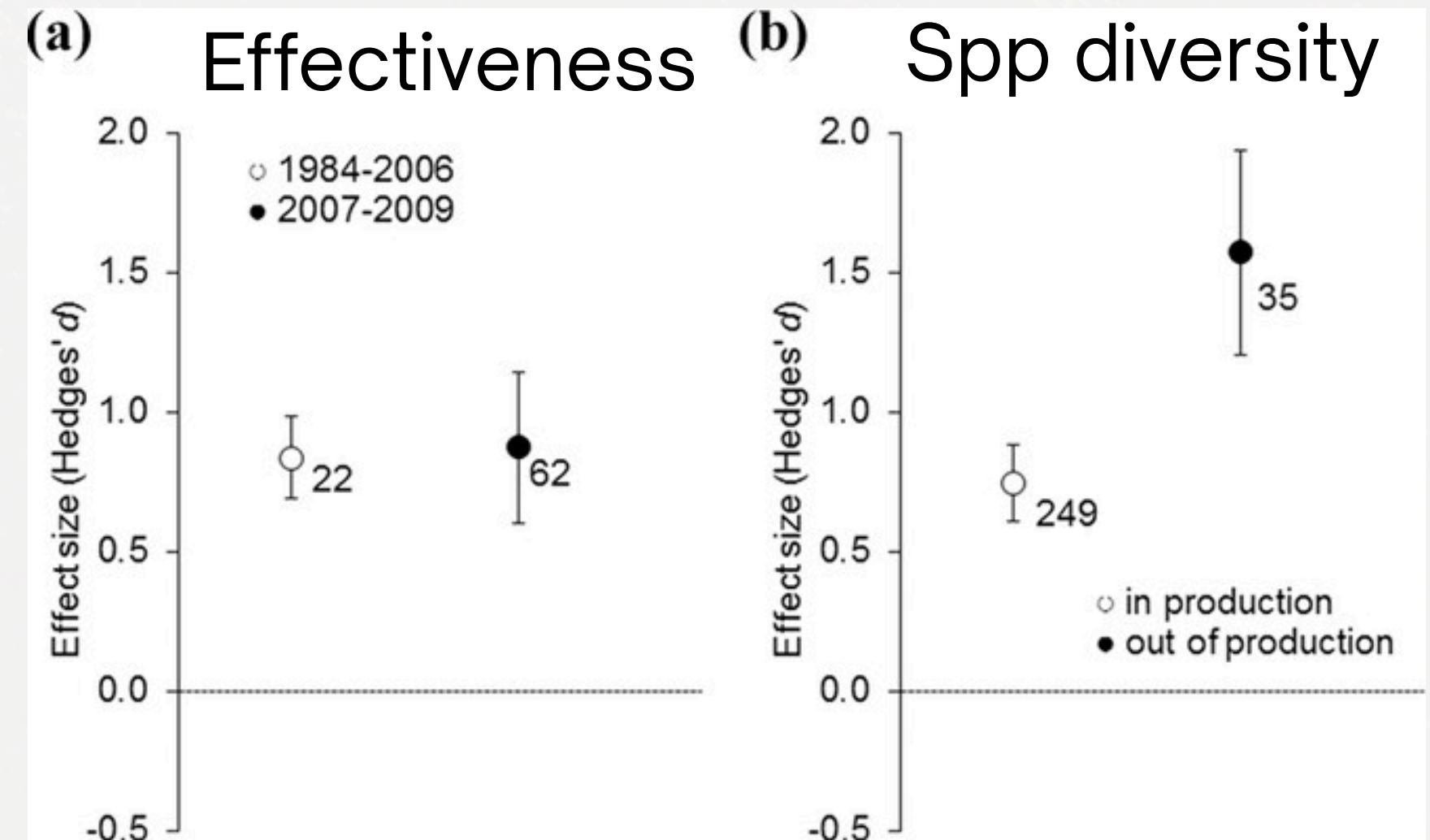
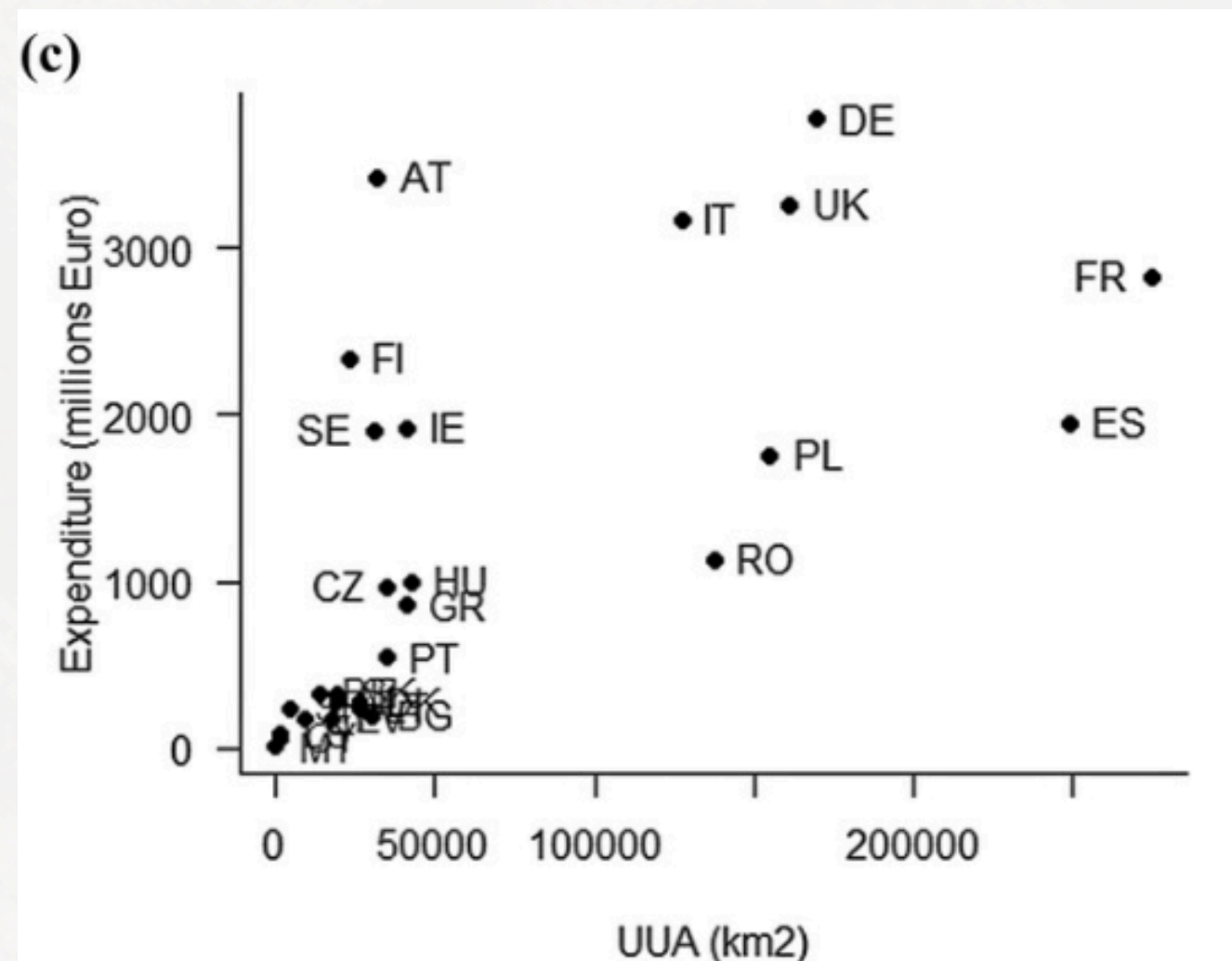
Richness vs evenness



Proportion of species
known vs thought to exist

How to protect biodiversity?

Many ways, little consensus, it is complicated, but...
agro-environmental schemes



Utilized agricultural area (UAA) relative to total realized expenditure

Very expensive

low effectiveness
(but equivocal evidence)

Baselining biodiversity

GBIF

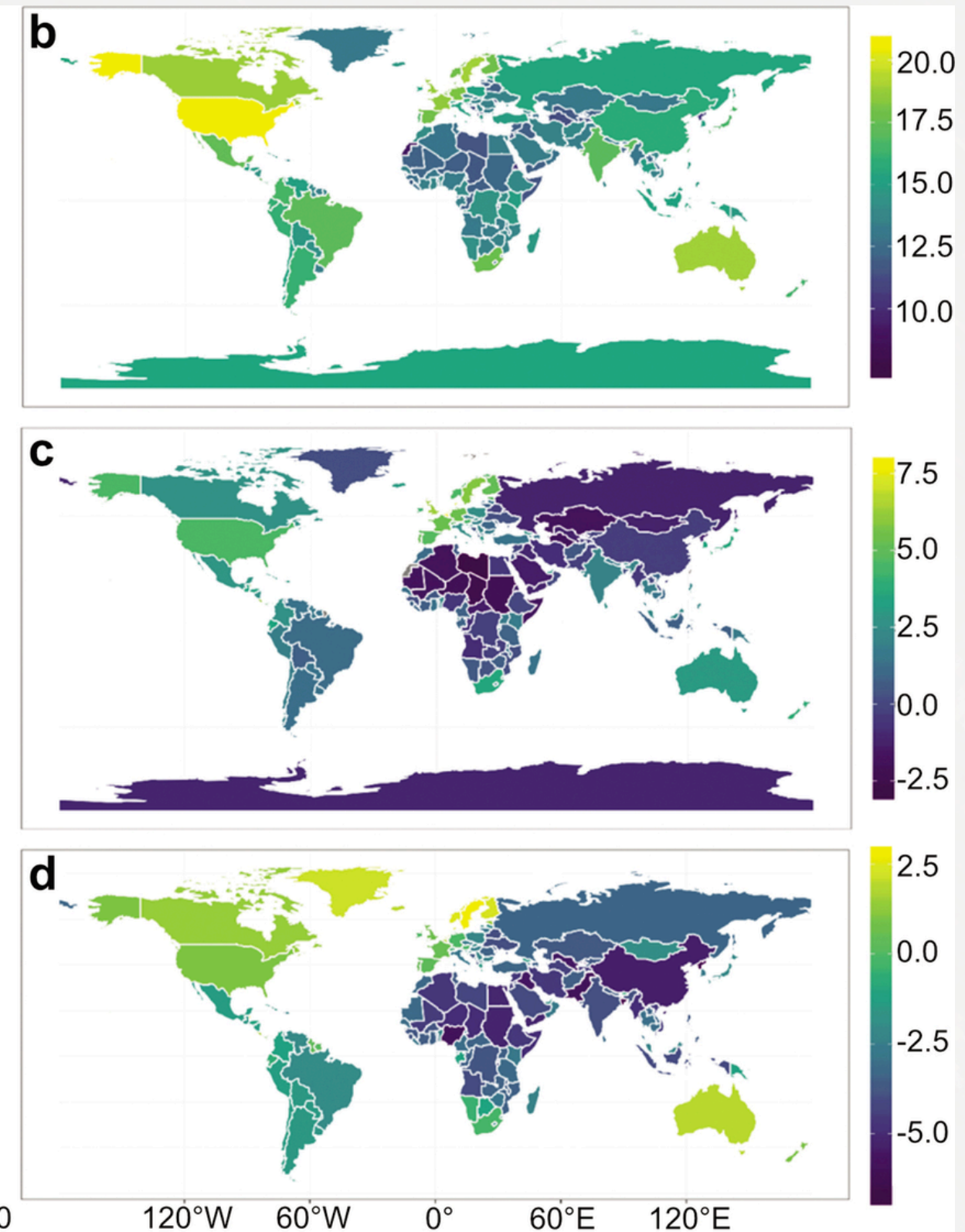
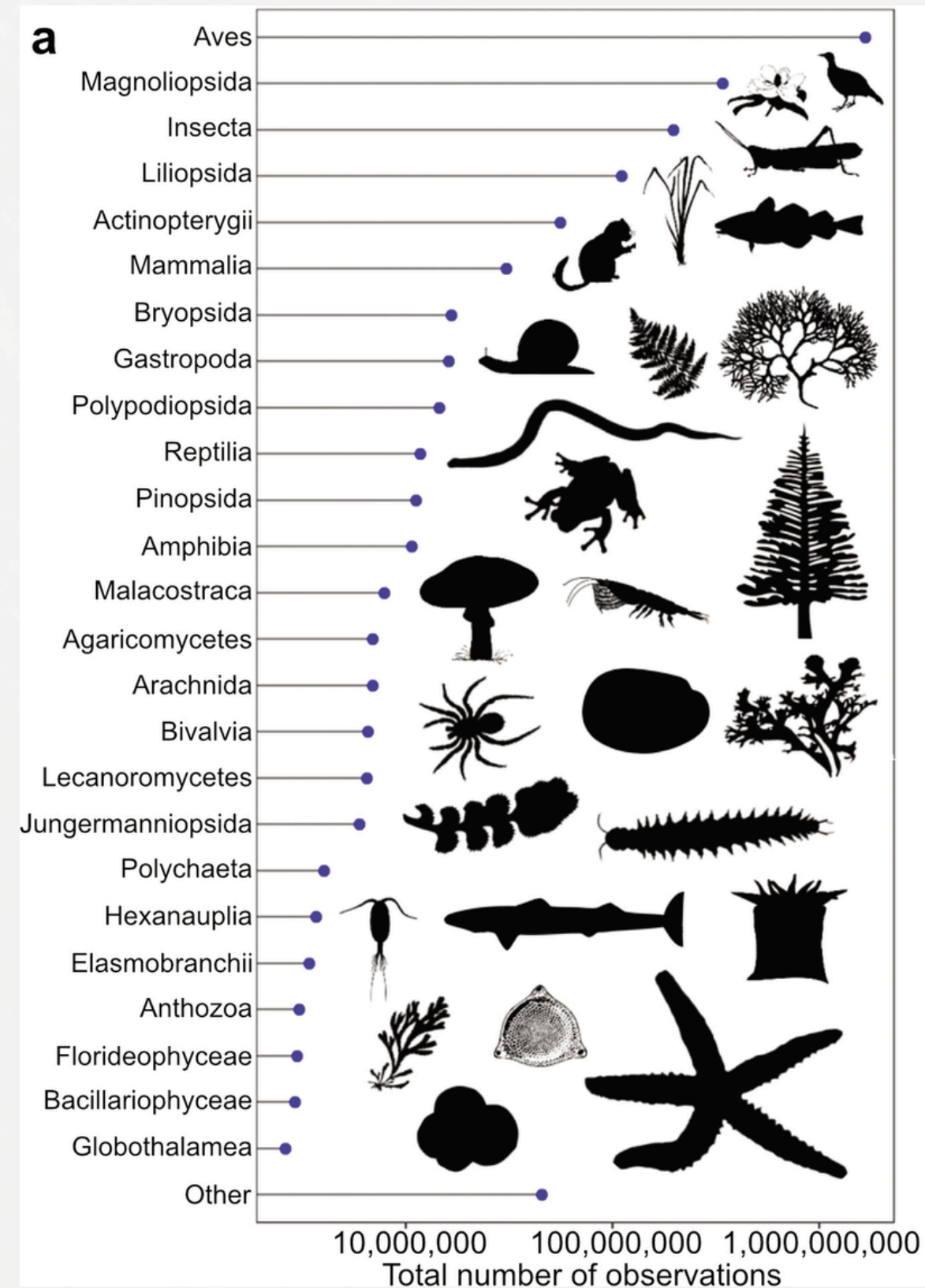
(Global Biodiversity Information Facility)

(a) # of records for the top 25 classes present in GBIF

(b) # of records in GBIF, by country, on a log scale

(c) # of records in GBIF, standardized by the area of the country

(d) The total number of records in GBIF per capita



JOURNAL ARTICLE

Three Frontiers for the Future of Biodiversity Research Using Citizen Science Data ^{FREE}

Corey T Callaghan ✉, Alistair G B Poore, Thomas Mesaglio, Angela T Moles, Shinichi Nakagawa, Christopher Roberts, Jodi J L Rowley, [Adriana Vergés](#), John H Wilshire, William K Cornwell

BioScience, Volume 71, Issue 1, January 2021, Pages 55–63,

GBIF Occurrences

Get data

How-to

Tools

Community

About

<

Occurrences

1

Year

▼

Month

▼

Location

^

No preference

Including coordinates

Without coordinates

Include records where coordinates are flagged as suspicious

+

-



© OSM, © OMT, GBIF.

SEARCH OCCURRENCES

215,018,127 RESULTS

TABLE

GALLERY

MAP

TAXONOMY

METRICS

↓

DOWNLOAD

	Scientific name	Country or area	Coordinates	Event date
	<i>Erannis defoliaria</i> (Clerck, 1759)	United Kingdom of Great ...	50.9N, 1.4W	2025 Jan
	<i>Operophtera brumata</i> (Linnaeus, 1758)	United Kingdom of Great ...	51.4N, 0.2W	2025 Jan
	<i>Erannis defoliaria</i> (Clerck, 1759)	United Kingdom of Great ...	50.9N, 1.4W	2025 Jan
	<i>Erannis defoliaria</i> (Clerck, 1759)	United Kingdom of Great ...	50.9N, 1.4W	2025 Jan
	<i>Operophtera brumata</i> (Linnaeus, 1758)	United Kingdom of Great ...	50.9N, 1.4W	2025 Jan
	<i>Phigalia pilosaria</i> (Denis & Schiffermüller) 1...	United Kingdom of Great ...	50.9N, 1.4W	2025 Jan
	<i>Pica pica</i> (Linnaeus, 1758)	United Kingdom of Great ...	53.4N, 2.3W	2025 Jan 01
	<i>Petricolaria pholadiformis</i> (Lamarck, 1818)	Belgium	51.1N, 2.5E	2025 Jan 03
	<i>Petricolaria pholadiformis</i> (Lamarck, 1818)	Belgium	51.1N, 2.7E	2025 Jan 02
	<i>Vespa velutina</i> Lepeletier, 1836	Belgium	51.1N, 2.6E	2025 Jan 02
	<i>Leptoglossus occidentalis</i> Heidemann, 1910	Belgium	51.1N, 2.6E	2025 Jan 03

GBIF Occurrences

☒

Aves

Year

Between

2000

2025

☒

Shropshire - GBR.1.84.1_1

Dataset

Search

☐

EOD – eBird Observation Dataset

195,669

☐

RSPB Big Garden Birdwatch winter sightin...

24,222

☐

Birda - Global Observation Dataset

15,371

☐

Bird Conservation Targeting Project (BCTP) i...

3,809

☐

iNaturalist Research-grade Observations

2,047

☐

Bird Conservation Targeting Project (BCTP) i...

1,717

☐

Montgomeryshire Wildlife Trust records hel...

1,362

☐

Miscellaneous records held on the Cofnod d...

1,241

☐

Miscellaneous records held by BIS

1,227

☐

DELTATRACK - Herring gulls (Larus argentat...

1,134

SEARCH OCCURRENCES | 250,965 RESULTS

TABLE

GALLERY

MAP

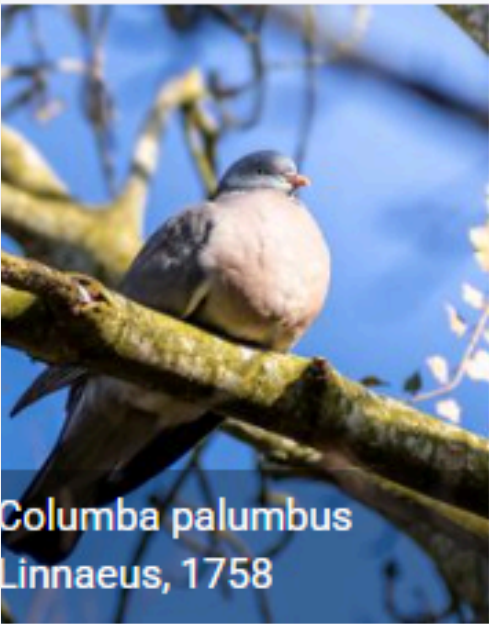
TAXONOMY

METRICS

DOWNLOAD

	Scientific name	Country or area	Coordinates	Event date
	<i>Passer domesticus</i> (Linnaeus, 1758)	United Kingdom of Great ...	52.7N, 2.8W	2025 Jan 05
	<i>Buteo buteo</i> (Linnaeus, 1758)	United Kingdom of Great ...	52.6N, 2.5W	2025 Jan 05
	<i>Spinus spinus</i> (Linnaeus, 1758)	United Kingdom of Great ...	52.5N, 3.2W	2025 Jan 11
	<i>Erithacus rubecula</i> (Linnaeus, 1758)	United Kingdom of Great ...	52.7N, 2.7W	2025 Jan 22
	<i>Milvus milvus</i> (Linnaeus, 1758)	United Kingdom of Great ...	52.7N, 2.3W	2025 Jan 26
	<i>Cyanistes caeruleus</i> (Linnaeus, 1758)	United Kingdom of Great ...	52.7N, 2.8W	2025 Jan 28
	<i>Bucephala clangula</i> (Linnaeus, 1758)	United Kingdom of Great ...	52.9N, 2.8W	2025 Jan 19
	<i>Bucephala clangula</i> (Linnaeus, 1758)	United Kingdom of Great ...	52.9N, 2.8W	2025 Jan 19
	<i>Accipiter nisus</i> (Linnaeus, 1758)	United Kingdom of Great ...	52.9N, 2.7W	2025 Jan 02
	<i>Passer domesticus</i> (Linnaeus, 1758)	United Kingdom of Great ...	52.7N, 2.7W	2025 Jan 25

GBIF Occurrences



Columba palumbus
Linnaeus, 1758



Egretta garzetta (Linnaeus, 1766)



Sitta europaea Linnaeus, 1758



Parus major Linnaeus, 1758



Turdus philomelos C.L.Brehm
1831



Falco tinnunculus Linnaeus, 1758



Prunella modularis (Linnaeus,
1758)



Turdus merula Linnaeus,
1758



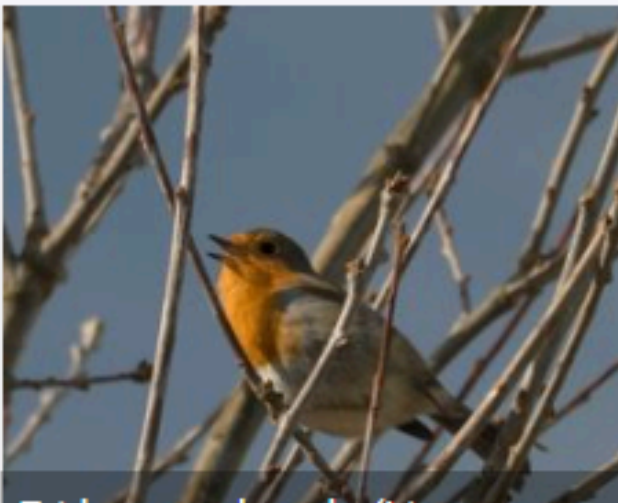
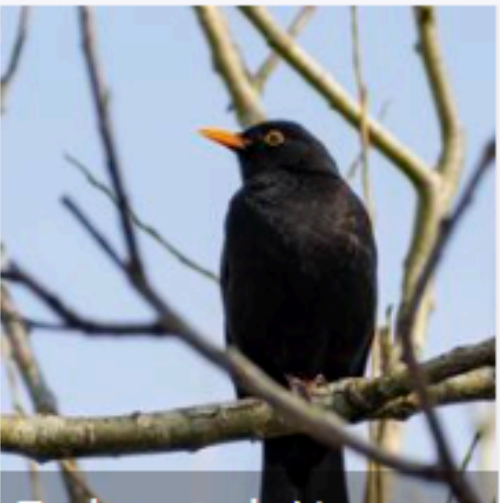
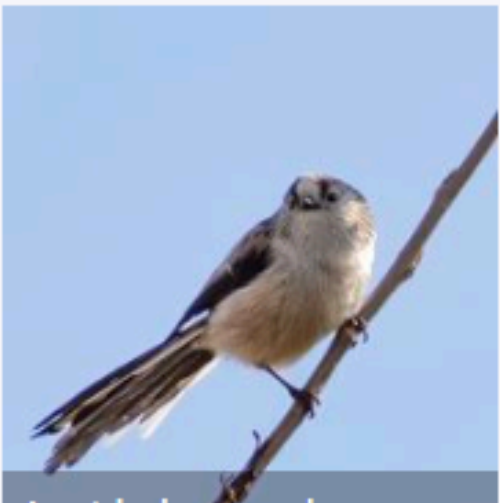
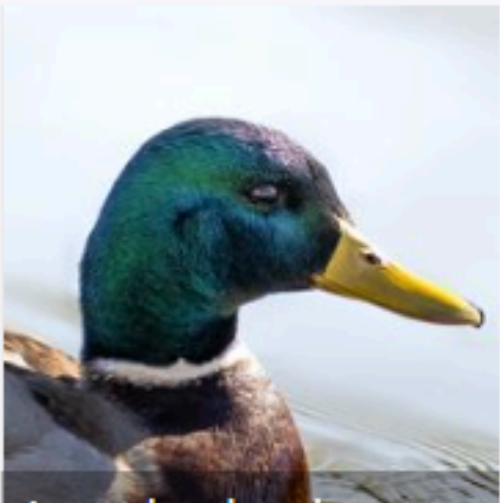
Corvus corone Linnaeus,
1758



Buteo buteo (Linnaeus, 1758)



Erithacus rubecula
(Linnaeus, 1758)



Code time

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